YOLO based Object Detection in Advanced Driver Assistance Systems -ADAS

M.Ayyadurai,
Department of Computer Science and Engineering
Rajalakshmi Engineering College
Chennai, India,
ayyadurai.m@rajalakshmi.edu.in

D.Anu ,
Department of Computer Science and Engineering
Presidency College, Hebbal,
Bangalore, India
anudevanmanjesh@gmail.com

Abdelhalim Mohammad Jubran,
Faculty of Economics and Administrative Sciences, Department of Economics, Zarqa University, College of Business Administration, University of Business and Technology, Jeddah ,Saudi Arabia,
ajubran@zu.edu.jo

Ankur Bahl,
School of Mechanical *Engineering, Lovely Professional University, India,*
ankur_bahl65@gmail.com

M Dinesh,
Department of VLSI Microelectronics, Saveetha School of Engineering, Saveetha Institute of Medical and Technical Sciences(SIMATS),Saveetha University, Chennai,
mdinesher@gmail.com

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***Abstract*— In recent years, there has been a boom in advanced driver assistance systems (ADAS) technology mainly to improve safety of driving. This technology aids in identifying hazardous driving conditions that may result in traffic collisions. Such incidents can arise from multiple causes, including driver distraction, excessive speed, and intoxication, poor visibility from fog, inadequate headlight illumination, or abrupt obstacles appearing on the roadway. The sophisticated capabilities of ADSA encompass parking support, road traffic recognition, object identification and lane rejection. Despite being an advanced technology, ADAS has specific limitations particularly in the speed and precision of object detection and tracking. While algorithms like R-CNN, Fast R-CNN and YOLO (You Only Look Once) are employed for object detection, they occasionally fail to identify objects with sufficient speed. To overcome this issue, the suggested work proposes a novel approach titled "Object Detection in Advanced Driver Assistance Systems (ADAS) Using YOLO". In this paper, a custom dataset is developed. This dataset is annotated and trained on YOLOv5, YOLOv7, YOLOv8. The outcome of all the models are compared based on mean average precision (mAP)and the results are evaluated to find the best model of ADAS.**

***Keywords*—Object detection, ADAS, CNN, R-CNN, YOLOV8, YOLO V5, YOLO V7.**

I. INTRODUCTION

Advanced Driver Assistance Systems (ADAS) is one of the most rapidly growing technologies when it comes to transportation. Over the past few years transportation safety and efficiency has become a very important aspect of transportation infrastructure. Even with a lot of advancement in the technology to improve road and driver safety, there are still a lot of deaths and accidents, approximately 1.19 million lives are lost annually due to road collision accidents worldwide [2]. One key reason for the accidents was the driver's lack of ability to focus on the road and be aware of the surroundings. With the involvement of ADAS, the driver can be alerted or the data from the object detection system can be used by the autonomous driverless systems. So to improve this as a key underlying technology for autonomous vehicles, a robust advanced driver assistance system (ADAS) is designed to automate, adapt, and improve automotive technology to promote safety and best driving experiences.

Object Detection can be done by using multiple different algorithms. Primarily the Convolutional-Neural-Network (CNN) is proven to be the best amongst the various recognition algorithms. While training such an deep learning model - CNN learns straight from the authentic pixel data [3]. That gives the access to explore complex contextual information and data features and also express them successfully. In this Paper under the CNN family, (You only look once) - YOLO is used for the detection of vehicles, pedestrians, potholes, etc. The YOLOv8[4] algorithm is chosen as the best fit for the application as a robust tool for object detection for improving the accuracy, speed and efficiency of the ADAS systems.

A. Problem Statement

The traditional method for training ADAS was by using cameras, ultrasonic sensors, radar and lidar, these are deployed to gather data about the vehicle's surroundings. The data undergoes various processing via algorithms built inside the computer systems. Features of ADAS such as adaptive cruise control, lane keeping assistance and efficient emergency braking are possible by using these data with the vehicle control systems. While there have been previous attempts at object detection using deep learning models like YOLOv5 andYOLOv7. They have proven to have a fair share of disadvantages.

YOLOv5 does not offer the same level of robustness when implemented in a diverse environment, and YOLOv7 performs less accurately and efficiently when compared to the latest iteration of the algorithms. This causes the ADAS system to be less accurate with its detection and takes up a lot of time to make the detection which is not feasible in the case of a moving vehicle that needs this data for safe transportation. Addressing these challenges is paramount for

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improving road safety, reducing vehicle damage, and maximizing road maintenance efforts.

II. RELATED WORKS

In the recent decades the technology has evolved and thriving from the timeline where lack of technology for object detection to so many architectures and methodology for object detection. Object detection has criteria which first decide on what you want to detect and in this research, we detect pedestrians, cars, trucks and potholes and train the dataset to the point of interest.

Region-based Convolutional Neural Network (R-CNN) was used before YOLO (You Only Look Once) and SSD (Single Shot Detector). R-CNN uses a selective search algorithm to extract 2000 regions from the input images based on the important features [5]. This model also uses a box fusion algorithm which, to eliminate the repetitive boxes in the image to get a better location of the final box in the image [6].

YOLOv4 Network is a model that is based on the previous YOLOv3 model. This model uses the CSPDarknet53 as its backbone network, improved from Darknet53 [7]. For object detection, one of the methods used was the YOLOv5, Scaled - YOLOv4 and YOLOR for detection and tracking on edge devices [8]. But this method lacked the speed and accuracy for a real world ADAS application in the road conditions in the present world.

Single Shot Multibox Detector [9] is similar to YOLO and offered a higher accuracy. It also works with 6 additional aspect ratios than YOLO. The high accuracy comes with the trade-off of speed. In an ADAS application with fast moving objects it calls for high-speed detection and such high-speed detection and accuracy can be achieved by YOLO.

III. METHODOLOGY

A. Architecture

The YOLOv8 [10] architecture is shown in fig.1. YOLOv8 is built on convolutional neural network, which are divided into two main parts: backbone and head. The CSPDarknet53 feature extractor, represents a significant improvement over previous backbone networks which are the YOLOv5 and YOLOv7. The combined features of the CSPDarknet53 and Cross-Stage Partial (CSP) [11] connectivity gives the architecture great extraction capabilities while concurrently maintaining efficiency. The YOLOv8 head consists of convolutional layers which are followed by many fully connected layers. These layers are responsible for predicting bounding boxes and class probabilities for objects detected in the image [12]. YOLOv8

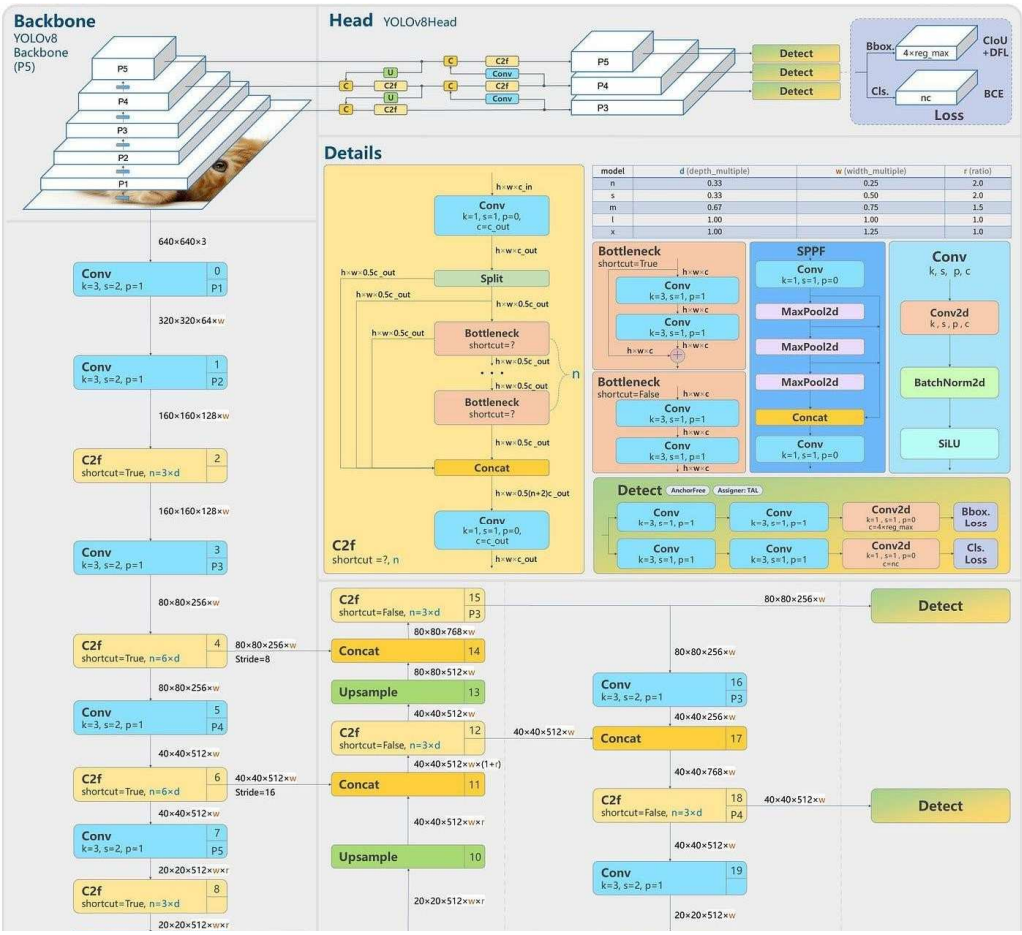

Fig.1. Architecture diagram of YOLOv8

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architecture replaces the original 6x6 convolutional layers with 3x3 convolutional layers for improved feature extraction. In the YOLOv8 [13], the detection head is the main component that processing the multidimensional features that are fetched from the backbone and generates predictions forobject detection. A detection head typically has a sequence of convolutional layers followed by a final output layer predicting bounding boxes, objectness values, and class probabilities. Components and features of YOLOv8 sensing head:

1. Convolutional layer: The convolutional layer works on multi-scale features that have been extracted from the backbone network. These layers help refine the features and generate higher level abstract representations which play a vital role in object detection.

2. Prediction layer. Prediction layer is the second layer of our network. It has to predict bounding box coordinates (x, y, width, height) and objectness values for each grid cell in the input image as well as produce a class probability for each class label.

4. Activation function: we use an activation function after the prediction layer— typically a sigmoid function. The purpose of this function is to scale all values produced by our model down into a fixed range between zero and one which will then be used directly as activations indicating predictions about bounding box coordinates.

5. Loss function: One of the elements that is involved in training the model using annotated data is the loss function. The model's prediction generated by the detection head should be compared with ground truth annotations during training. A particular loss function used in this comparison can measure how wrong the model's prediction is to help it learn from its mistake and improve its predictions over time.

6. Post - Processing: once the sensing head provides its predictions, specific post-processing techniques should be considered as an attempt to enhance these results.

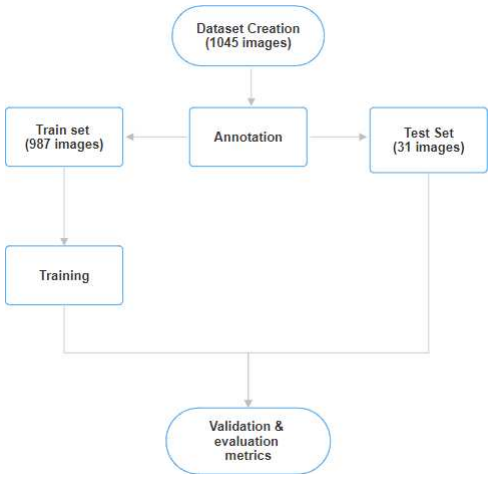

Fig.2. Flow Diagram

B. *Dataset*

The dataset used for the object detection are sourced from Robo Flow Universe, specifically the “Autopilot Dataset” by Arpan Singh [14] and “Pothole Detection Yolo v8 Dataset” by Kartik [15]. A dataset from Kaggle was sourced for the pedestrian images. The dataset has totally 1045 images in that 987 images are used in the training phase, 27 images are managed for the validating our data set and 31 images utilized for the testing set. As shown in the flow diagram Fig.2.

Both these datasets were combined to create a custom dataset that includes classes such as car, truck, pedestrian and potholes as in fig.3. The dataset contains images captured from the dashboard of a car on a highway. It also includes images which contain potholes, cars and pedestrians in a diverse environment under different lighting conditions and angles. These images were annotated using labelimg software as shown in fig.4.

Fig.3. Images of the newly created dataset

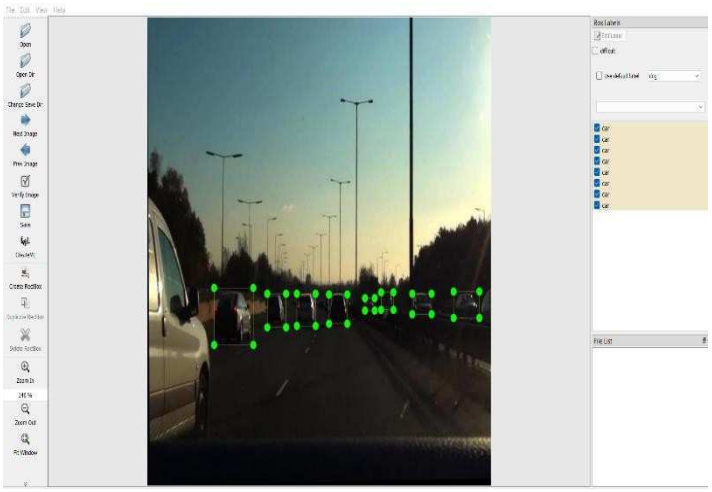

Fig. 4. Annotation of an image using Labelling tool

C. *Preprocessing*

To enhance our custom data model, the dataset underwent several preprocessing methods. This ensures the robustness of the model. All the images were oriented and were resized to 640 x 640 pixels.

Data augmentation techniques were performed to further diversify the dataset. The following modifications were done:

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1. Exposure: The exposure was changed between +15% to -15%.
2. Shear: A horizontal and vertical shear was added to the images.

D. Model Training

Data Collection from Real-World Driving Scenarios:

Purpose: The main purpose is to ensure YOLOv8 models effectiveness in practical scenarios that initially gather the dataset in a diverse way to accurately represent real world driving conditions.

Approach: The dataset which we collect is a diverse dataset as it contains urban, highway and rural environments and it is done using the help of vehicle mounted sensors such as camera, lidar, radar and GPS.

Annotation: Here annotating by manually marking points of interest from the collected dataset such as vehicles, pedestrians, potholes etc. This annotation process is very important for training the model for object detection which results in accurate recognition.

Diversity: The dataset collected has a wide range of factors such as includes different lighting conditions (day and night) weather conditions (rain, fog, sunny) road types (highways and city roads) and traffic conditions (intersections, Parking)

Augmentation: To increase the reliability the data augmentation techniques are used. These include transformations of images such as rotating, scaling, cropping, flipping, and adjusting brightness and contrast for the algorithm to analyze.

Training the YOLOv8 Model with Annotated Datasets: The model was trained using python under Google Colab.

Model Selection: Choose the appropriate YOLOv8 architecture option for your task, considering factors such as input resolution, backbone network (e.g. CSPDarknet53),and hyperparameters. Choosing the appropriate architecture for the task

Regularization: Applying regularization techniques such as batch normalization, weight reduction to mainly prevent overfitting and improve the generalization performance.

Fine-Tuning: Fine-tune the model for a specific task or dataset as needed to further improve performance.

Setting up the Evaluation Metrics for Accuracy:

Metric Selection: Select appropriate evaluation metrics to evaluate the accuracy and performance of YOLOv8 model for object detection task. The metrics included mean average precision (mAP), precision, recall.

Evaluation Procedure: Evaluating the trained YOLOv8 model using a separate validation dataset to quantify the model's performance by calculating selected evaluation metrics.

Analysis and Interpretation: Analyze the evaluated results to identify performance metrics and areas for improvement in the YOLOv8 model. This analysis is critical for making conclusions and recommendations for future improvements to the model.

IV. RESULTS

TABLE I. PERFORMANCE MATRIC

Model	**mAP@0.50**	**mAP@0.95**	**Precision**	**Recall**
YOLOv5 small	0.72	0.28	0.80	0.62
YOLOv7	0.67	0.32	0.74	0.65
YOLOv8 nano	0.76	0.37	0.85	0.70

TABLE II. CLASS-WISE MEAN AVERAGE PRECISION (MAP) FOR YOLOV8 NANO

Class	**mAP@0.50**	**mAP@0.95**
Overall	0.76	0.37
Car	0.84	0.42
Pedestrian	0.99	0.47
Pothole	0.55	0.26
Truck	0.66	0.34

Table I shows the results obtained by training our custom dataset on the YOLO architectures for epochs 100. In the evaluation, the main metric is mAP, precision and recall.

Table II shows the results for each of the class for the YOLOv8 nano model. Our model has attained a mAP of 0.76 at 0.5 IoU, mAP of 0.37 at 0.95 IoU, a precision(P) of 0.85 and a recall (R) of 0.70. The confusion matrix for the trained model is shown in Fig.5 illustrates the true positive (TP), true negatives (TN), false positive (FP), and false negatives (FN).

Comparing our YOLOv8 nano implementation with the previous works done, our model exhibits superior performance in all the performance metrics such as mAP, precision and recall. A sample of the detection result is shown in fig.6.

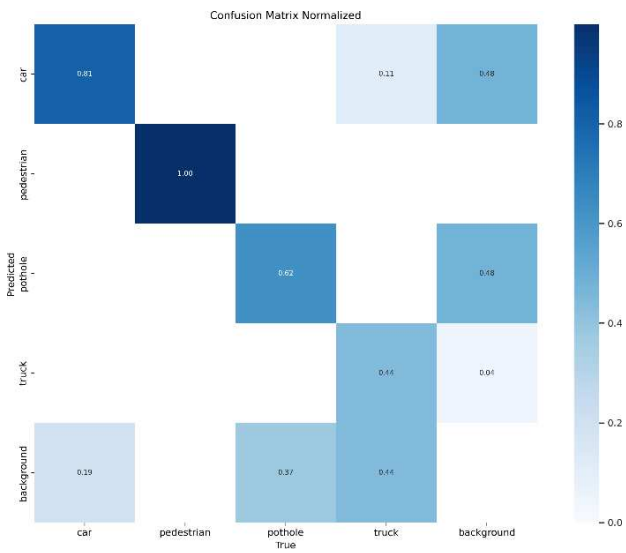

Fig.5. Confusion Matrix

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Fig.6. Some sample detection results

V. CONCLUSION

This research work evaluates the performance of YOLOv8 in the field of object detection model in the task of car, pedestrian, pothole and truck detection. Achieving an accurate predicting model is difficult due to the different classes in the dataset. Throughout this study, the custom dataset is trained using various versions of YOLO. The YOLOv8 model emerges as a robust and highly accurate model for object detection. This model is ideal in terms of striking a equalization between the accuracy, efficiency and speed. In the future, we can implement this model in real-world applications and further improve and optimize the performance of the model for different environments and conditions.

VI. REFERENCES

[1] K. Bengler, K. Dietmayer, B. Farber, M. Maurer, C. Stiller, and H. Winner, “Three decades of driver assistance systems: Review and future perspectives,” IEEE Intell. Transp. Syst. Mag., vol. 6, no. 4, pp. 6–22, 2014.

[2] “Road traffic injuries: A major global public health crisis,” Public Health, vol. 122, no. 12, pp. 1399–1406, Dec. 2008.

[3] R. Girshick, “Fast R-CNN,” Apr. 30, 2015. Accessed: May 05, 2024. [Online]. Available: http://arxiv.org/abs/1504.08083

[4] D. Reis, J. Kupec, J. Hong, and A. Daoudi, “Real-Time Flying Object Detection with YOLOv8,” May 17, 2023. Accessed: May 05, 2024. [Online]. Available: http://arxiv.org/abs/2305.09972

[5] R. Girshick, J. Donahue, T. Darrell, and J. Malik, “Region-Based Convolutional Networks for Accurate Object Detection and Segmentation,” IEEE Trans. Pattern Anal. Mach. Intell., vol. 38, no. 1, pp. 142–158, Jan. 2016

[6] N. Duraimurugan, R. A and R. S, "ResQ – Intelligent Real-Time Disaster Management Using Machine Learning and IoT," 2024 International Conference on Innovative Computing, Intelligent Communication and Smart Electrical Systems (ICSES), Chennai, India, 2024, pp. 1-7, doi: 10.1109/ICSES63760.2024.10910703.

[7] P. Wang, X. Wang, Y. Liu, and J. Song, “Research on road object detection model based on YOLOv4 of autonomous vehicle,” IEEE Access, vol. 12, pp. 8198–8206, 2024.

[8] Kumar P., Lalitha, H., Priya, D.J. et al. (2025), AI-driven plant health monitoring: evaluating the WRLSB-HPS algorithm for leaf disease classification. Earth Sci Inform 18, 288 (2025). https://doi.org/10.1007/s12145-025-01762-8 SCI

[9] P. Kumar, H. M and H. V (2025), "Development of Interactive Assistance for Academic Preparation Using Large Models Language," 2025 International Conference on Computational, Communication and Information Technology (ICCCIT), Indore, India, 2025, pp. 265-269, doi: 10.1109/ICCCIT62592.2025.10928137. https://ieeexplore.ieee.org/document/10928137

[10] H. Wang, C. Liu, Y. Cai, L. Chen, and Y. Li, “YOLOv8-QSD: An improved small object detection algorithm for autonomous vehicles based on YOLOv8,” IEEE Trans. Instrum. Meas., vol. 73, pp. 1–16, 2024.

[11] J. Terven, D.-M. Córdova-Esparza, and J.-A. Romero-González, “A Comprehensive Review of YOLO Architectures in Computer Vision: From YOLOv1 to YOLOv8 and YOLO-NAS,” Machine Learning and Knowledge Extraction, vol. 5, no. 4, pp. 1680–1716, Nov. 2023.

[12] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, “You Only Look Once: Unified, Real-Time Object Detection,” Jun. 08, 2015. Accessed: May 05, 2024. [Online]. Available: http://arxiv.org/abs/1506.02640

[13] N. Duraimurugan, V. S. S M and S. S, "Deep Learning - Driven RPA Bot for Automated Evaluation of Handwritten Examination Papers Using Gen AI," 2024 International Conference on Innovative Computing, Intelligent Communication and Smart Electrical Systems (ICSES), Chennai, India, 2024, pp. 1-5, doi: 10.1109/ICSES63760.2024.10910366

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